Brieuc Collet, Researcher in Space Physics

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Space physics researcher specialized in planetary wave-plasma instability

Employment History

2025– 2027 **Post Doc** PPARC, Tohoku University, Japan
Understanding Mercury and the Jovian environment with in situ radio measurements from Juno, Bepi and Juice

2022– 2025 **PhD in Astrophysics** Lab of Astrophysics of Marseille, Aix-Marseille University, France
Thesis title : *Understanding the microphysics of Jovian radio auroral emissions with Juno space probe*

2022 **Master 2 Internship** Laboratory of Astrophysics of Marseille (France)
Subject : *Jovian radio auroral emissions : preliminary works to my PhD*

2021 **Master 1 Internship** LESIA, Observatory of Paris (France)
Subject : *Comparative study of auroral radio sources from Jupiter and Saturn*

Education

2022 – 2025 **PhD in Astrophysics, Aix-Marseille Université**

2021 – 2022 **Master's Degree Plasma Physics, Paris Saclay University**

2019–2020 **Bachelor's Degree of Fundamental Physics, Paris Saclay University**

Teachings

2022-2025 **Teaching Assistant**, 192h, Aix-Marseille University

Scientific Outreach

- 2022–2025 **Festival of Astronomy of Provence**
- 2022–2025 **Fête de la Science - French Science Diffusion Day**

Skills

Code Proficiency in **Python**, Knowledge in **C++** and **IDL**

Astronomy Observation nights at Observatory of Haute Provence

Community Activities

2022–2025 **Laboratory's Sustainable Development Group Member**
Laboratory's Seminar Organizing Committee

2023–2025 **PhD Representative in Laboratory Council**

2025 **Local Organizing Committee** for PREX, Marseille France
Reviewer for PREX proceedings

Awards

2022 PhD grant

2025 **Best PhD presentation award**, Planetary, solar and heliospheric Radio Emissions X, Marseille, France

Publications

Articles

[1] Al Saati, S., Clément, N., […], **Collet, B.** et al. 2022. « Magnetosphere -Ionosphere -Thermosphere Coupling Study at Jupiter Based on Juno's First 30 Orbits and Modeling Tools ». In : *Journal of Geophysical Research : Space Physics*. DOI : 10.1029/2022JA030586.

[2] Benmahi, B., Hue, V., […], **Collet, B.** et al. 2025. « Precipitating electron properties of Ganymede's aurora retrieved from Juno/UVS observations during PJ34 ». In : *Astronomy & Astrophysics*. DOI : 10.1051/0004-6361/202556996.

[3] **Collet, B.**, Lamy, L. et al. 2023. « Characterization of Jovian hectometric sources with Juno : statistical position and generation by shell-type electrons ». In : *PRE IX*. Sous la dir. de C. K. Louis, C. M. Jackman et al. DIAS et TCD. DOI : 10.25546/103095.

[4] **Collet, B.**, Lamy, L. et al. 2024. « A New Type of Jovian Hectometric Radiation Powered by Monoenergetic Electron Beams ». In : *Journal of Geophysical Research : Space Physics*. DOI : 10.1029/2024JA032422.

[5] **Collet, B.**, Lamy, L. et al. 2025. « In Situ Analysis of Jupiter's Broadband Kilometric Auroral Radio Emissions With Juno ». In : *Geophysical Research Letters* 52.9. DOI : 10.1029/2024gl114447.

[6] **Collet, B.**, Lamy, L. et al. 2026. « Detection of Jovian kilometric and hectometric auroral radio harmonics with Juno/Waves in situ measurements ». In : *Planetary, Solar and Heliospheric Radio Emissions X*. Sous la dir. de L. Lamy, C. K. Louis et al. Accepted, preprint. OSU Pythéas/AMU, Observatoire de Paris. DOI : 10.25935/PREX-TR41.

[7] Heidari, N., Boisse, I., […], **Collet, B.** et al. 2024. « The SOPHIE search for northern extrasolar planets : XIX. A system including a cold sub-Neptune potentially transiting a V = 6.5 star HD 88986 ». In : *Astronomy & Astrophysics*. DOI : 10.1051/0004-6361/202347897.

[8] Lamy, L., Cecconi, B., […], **Collet, B.** et al. 2023. « Re-exploring the radio spectrum of Uranus in orbit : science case and digital high-frequency receiver ». In : *PRE IX*. Sous la dir. de C. K. Louis, C. M. Jackman et al. DIAS et TCD. DOI : 10.25546/103105.

[9] Louis, C. K., Louarn, P., **Collet, B.** et al. 2023. « Source of Radio Emissions Induced by the Galilean Moons Io, Europa and Ganymede : In Situ Measurements by Juno ». In : *Journal of Geophysical Research : Space Physics*. DOI : 10.1029/2023JA031985.

[10] Louis, C.K., Boudouma, A., […], **Collet, B.** et al. 2026. « Search for the influence of the Galilean moons on Jovian kilometric and decametric radio emissions over 7 years of Juno data ». under review.

[11] Rabia, J., Hue, V., […], **Collet, B.** et al. 2025. « In situ and remote observations of the ultraviolet footprint of the moon Callisto by the Juno spacecraft ». In : *Nature Communications*. DOI : 10.1038/s41467-025-62520-4.

[12] Wu, S., Whiter, D., […], **Collet, B.** et al. « A radio oval above Earth's auroral oval ». Under review, Science Advances.

Thesis

[13] **Collet, B.** 2025. « Comprendre la microphysique des émissions radio aurorales de Jupiter avec Juno ». Thèse de doctorat dirigée par Lamy, L. au Laboratoire d'Astrophysique de Marseille, Aix-Marseille Université. Thèse de doct.

Conferences

Orals

2022 **Planetary, solar and heliospheric Radio Emissions IX**, Dublin, Irlande

2023 **Chapman Understanding Alfvén Waves in the Sun and the Heliosphere**, Berlin, Allemagne

2024 **Coloque national de prospectives (PNST)**

Magnetospheres of Outer Planets, Minneapolis, USA

2025 **Planetary, solar and heliospheric Radio Emissions X**, Marseille, France

2026 **JPGU**, Chiba, Japan (next May)

Magnetospheres of Outer Planets, Toulouse, France (next July)

Posters

2022 **Coloque national de prospectives (PNST)**, Marseille, France

2024 **Astroradiofr24, French radio frequency astrophysics, towards SKA**, Paris

2025 **EGU General Assembly**, Vienne, Autriche